Menstrual pads and liners: HMTc Standard

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CORRESPONDENCE

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KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

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Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for menstrual pads, panty liners, intralabial and reusable pads (Feminine Care), evaluated on the total heavy-metal content of the finished article as placed on market (mucosal basis). No government sets a per-metal numeric content limit for this menstrual product, so every analyte is held to a disclosed read-across by analogy, the strictest health-based limit a government sets for an article in comparable prolonged mucosal contact, with inorganic arsenic and methylmercury bounded by their parent totals.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit default). Device forms carry no per-metal numeric heavy-metal content limit, so every panel cell is a disclosed read-across by analogy to the strictest health-based limit a government sets for an article in comparable prolonged mucosal contact; no occurrence percentile is computed and no number is originated by HMTc. Cosmetic forms read across the strictest sovereign mucosal-contact cosmetic contaminant limit. Two basis families (device content, cosmetic content) are never conflated, and neither is a leachate/migration number. Provenance vocabulary: {regulatory-alignment, read-across, not-material, control, ratchet}.

2 | Master limit table

All values in ppb (µg/kg), as sold basis. A brand meeting every value qualifies for the mark.

ANALYTE	HMTC LIMIT	BASIS

Lead (Pb)	500	human-grade read-across (DE)
Cadmium (Cd)	100	human-grade read-across (DE)
Total arsenic (tAs)	500	human-grade read-across (DE)
Inorganic arsenic (iAs)	≤ tAs	bounded by tAs (reflex speciation)
Mercury, total (tHg)	100	human-grade read-across (DE)
Methylmercury (MeHg)	≤ tHg	bounded by tHg (reflex speciation)
Nickel (Ni)	10,000	human-grade read-across (South Korea)
Tin (Sn)	3,750,000	human-grade read-across (European Union)
Aluminium (Al)	560,000	human-grade read-across (European Union)
Total chromium (Cr)	50	human-grade read-across (AU)
Hexavalent chromium (Cr-VI)	5	human-grade read-across (European Union)

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	500	DE — human-food analog
Cadmium (Cd)	100	DE — human-food analog
Total arsenic (tAs)	500	DE — human-food analog
Inorganic arsenic (iAs)	≤ tAs	iAs carries no product-specific sovereign cell; it is a component of and bounded by its parent total tAs (iAs <= tAs), an enforceable ceiling via reflex speciation rather than an empty cell.
Mercury, total (tHg)	100	DE — human-food analog
Methylmercury (MeHg)	≤ tHg	MeHg carries no product-specific sovereign cell; it is a component of and bounded by its parent total tHg (MeHg <= tHg), an enforceable ceiling via reflex speciation rather than an empty cell.

Nickel (Ni)	10,000	South Korea — human-food analog
Tin (Sn)	3,750,000	European Union — human-food analog
Aluminium (Al)	560,000	European Union — human-food analog
Total chromium (Cr)	50	AU — human-food analog
Hexavalent chromium (Cr-VI)	5	European Union — human-food analog

4 | Cite this standard

Pendergrass, K. (2026). *Menstrual pads and liners: HMTc standard* (Standard No. 10.83138/hmtc.std.15.menstrual-pads). Institute of Contaminant Standards. https://doi.org/10.83138/hmtc.std.15.menstrual-pads

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References

[1] AU — human-food maximum levels
Binds Cr. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] DE — human-food maximum levels
Binds Pb, Cd, tAs, tHg. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[3] European Union — human-food maximum levels
Binds Sn, Al, Cr-VI. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[4] South Korea — human-food maximum levels
Binds Ni. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

STATUS AND LICENCE

A normative certification standard, editorially reviewed and version-controlled in-house; every value traces to a named sovereign maximum level, disclosed in the derivation. © 2026 Institute of Contaminant Standards / Heavy Metal Certified. All rights reserved.

COMPETING INTERESTS

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DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at

PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/menstrual-pads.

https://heavymetalcertified.com/standards/menstrual-pads. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

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